

Math 8 End of Q4 Test — Answer Key

Mary, Star of the Sea School · Dr. Park · Answer Key (Teacher Use)

1. $V = 20 \text{ V}$, $R = 5 \Omega$. I ? **Answer:** 4 A
2. $I = 3 \text{ A}$, $R = 10 \Omega$. V ? **Answer:** 30 V
3. $v = f\lambda$: $f = 10 \text{ Hz}$, $\lambda = 3 \text{ m}$. v ? **Answer:** 30 m/s
4. $v = 300 \text{ m/s}$, $f = 50 \text{ Hz}$. λ ? **Answer:** 6 m
5. If λ halves, f (constant v) ____? **Answer:** doubles
6. Kepler: r triples. T changes by factor of $\sqrt[3]{27} \approx ?$ **Answer:** 5.20
7. Earth $T = 1 \text{ yr}$. Planet at $r = 9 \text{ AU}$. T (yrs)? **Answer:** 27
8. Cylinder $r = 4$, $h = 6$, $\pi = 3.14$. V ? **Answer:** 301.44
9. Cone $r = 6$, $h = 4$, $\pi = 3.14$. V ? **Answer:** 150.72
10. Sphere $r = 6$, $\pi = 3.14$. V ? **Answer:** 904.32
11. Two-way table: 25 in row A, 35 in row B. Total? **Answer:** 60
12. Table shows 80 total, 30 in cat X. Cat Y? **Answer:** 50
13. $P = VI$: $V = 20$, $I = 4$. Power? **Answer:** 80 W
14. 650 nm to meters ($1 \text{ nm} = 10^{-9} \text{ m}$)? **Answer:** 6.5×10^{-7}
15. Pyramid $V = \frac{1}{3}Bh$. $B = 4 \times 5$, $h = 12$. V ? **Answer:** 80
16. Cone $r = 5$, $h = 12$, $\pi = 3.14$. V ? **Answer:** 314
17. Kepler: closer orbit = shorter period. True? **Answer:** True
18. 60 W bulb for 5 hrs. Energy? **Answer:** 300 Wh
19. Two-way tables show what kind of variables? **Answer:** Categorical
20. Cube side 8. Volume? **Answer:** 512